Python Encapsulation Interview Notes

1. Encapsulation

Encapsulation means bundling data (attributes) and methods into a single class while controlling access to the data.

2. _variable vs __variable

_variable (single underscore)
• Protected by convention.
• No special restriction by Python.
• Can be accessed directly.

__variable (double underscore)
• Uses name mangling.
• Internally renamed to _ClassName__variable.
• Prevents accidental access and avoids name conflicts in inheritance.
• Not truly private.

3. Java vs Python

Java
private variables are enforced by the compiler and cannot be accessed directly.

Python
No true private variables. __variable uses name mangling and can still be accessed with _ClassName__variable.

4. Interview Answers

Q: Difference between _variable and __variable?
_single underscore is protected by convention and directly accessible. Double underscore uses name mangling, making direct access difficult but not impossible.

Q: Is __variable truly private?
No. It is name-mangled, not truly private.

Q: Why use __?
To discourage accidental access and avoid name conflicts in inheritance.